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Design, Fabrication, Measurement, and Analysis of Photonic Circuits

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1 Introduction

Photonics is a growing field of technology, which offers many promising applications. Some of these include ultra fast data transmission, quantum computing, biosensing, and optics. As of 2026, silicon is often used in photonic chip manufacture because there is an existing infrastructure and methodology for the manufacture of silicon chips, and because a layer of silicon encased in an insulating layer of cladding, such as silicon dioxide, allows for a useful level of light information transmission. Light has different polarization modes, and TE and TM modes are useful in different applications.

1.1

2 Theory

In the manufacturing process, a layer of Si is deposited on top of a layer of SiO_2, and lithography is used to turn the Si layer into a thin waveguide. Another layer of SiO_2 is deposited after this, so that the Si waveguide is encased in an SiO_2 cladding.

The dimensions of the SiO_2 waveguide determine the effective index (n_{eff}) and group index (n_g) for the waveguide. The following figure of electric field intensity for a cross section of a 500 nm by 220 nm waveguide was obtained using Lumerical Mode.

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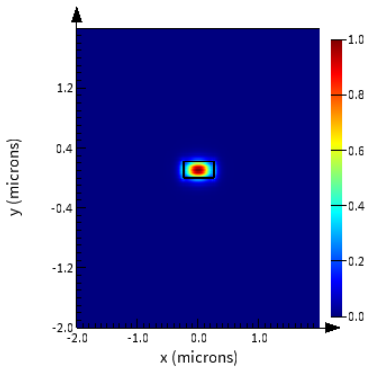

Fig. 1 Electric field intensity for the TE mode in a basic slab waveguide

By simulating wavelength dimensions in a range close to our design, we can obtain an idea of how possible manufacturing variations will effect waveguide transmission. This method is known as corner analysis.

Cross section (nm^2)	Fundamental quasi-TE	Fundamental quasi
510 x 215.3	4.18	3.58
510 x 223.1	4.18	3.84
500 x 220	4.2	3.71
470 x 215.3	4.25	3.48
470 x 223.1	4.26	3.74

Table 1 Corner analysis results for a 500 x 220 nm waveguide

The above results demonstrate that TE polarization is more stable and more sensitive to the width of the waveguide, while TM polarization is sensitive to both.

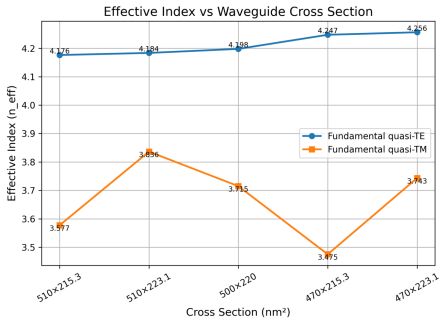

Fig. 2 Plot of corner analysis simulation data

A basic Mach-Zehnder interferometer is able to manipulate the free spectral range (FSR) of a light transmission as follows. The transfer function for an MZI is given by

$\frac{Io}{Ii} = \frac{1}{2}[1+\cos(b[?]L)]$

where I is the intensity and β is the phase constant, which is defined as:

$b = \frac{2\pi(neff)}{\lambda}$

The group index (n_g) is given by:

$n_g = n_{eff} - l * (dn_{eff}/dl)$

And the free spectral range (FSR) is given by:

$FSR = l^2/(n_g * DL)$

Therefore the spacing between adjacent peaks of the transmission through an MZI can be controlled by the path length difference of its varying arms. The figure below shows simulated gain peaks for an MZI.

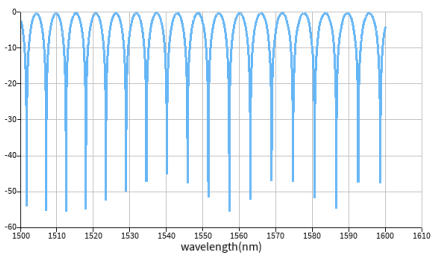

Fig. 3 Wavelength vs Gain in a simulated inteferometer

3 Simulation

Layout designs were created with the software Klayout. The figure below consists of two TE mode MZIs, consisting of two gratings, two Y branch beam-splitters, and two waveguide paths of varying lengths. The waveguide paths have a width of 500 nm. The first waveguide has path lengths of 180 and 280 micrometers, for a ΔL of 100 micrometers. The second waveguide has path lengths of 180 and 380 micrometers, for a ΔL of 200 micrometers.

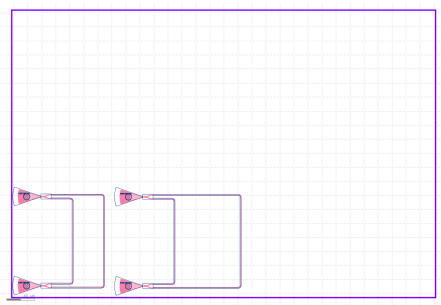

Fig. 4 First design draft 2/2/2026

The following two figures demonstrate how the difference in ΔL changes the transmission vs wavelength pattern between the two interferometer designs, with a larger ΔL resulting in a smaller FSR .

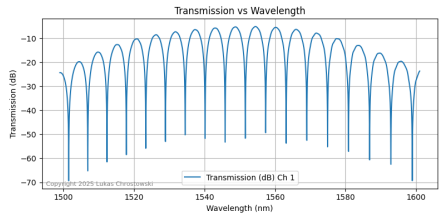

Fig. 5 Transmission vs Wavelength for 1000 nm [?]L

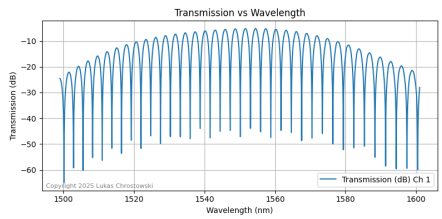

Fig. 6 Transmission vs Wavelength for 2000 nm [?]L

The following figure shows that greater wavelengths of light lead to a lower effective index, which results in a higher phase velocity for the light due to the equation β = 2πn/ λ

4 Manufacturing Results

Let us first compare data from these two MZIs using the Iris method, which gives a slightly higher R^2 value for MZI 1.

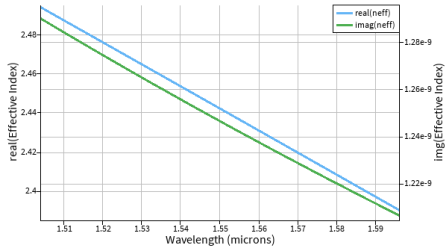

Fig. 7 neff vs wavelength

5

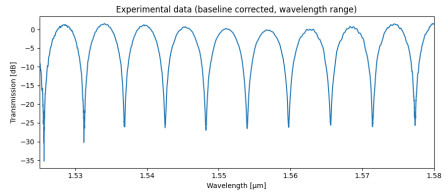

Fig. 8 MZI 1 baseline corrected

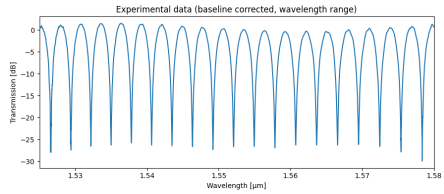

Fig. 9 MZI 2 baseline corrected

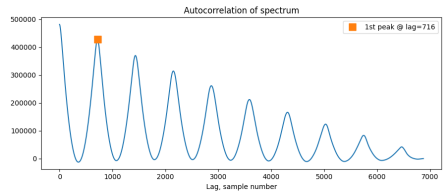

Fig. 10 MZI 1

As expected, the longer arm length in MZI 2 resulted in a lower FSR, with peaks tighter and closer together. Using the provided scripts, we create a mathematical model of the transmission data and shift it to match the peaks of our experimental data.

For both MZIs 1 and 2, we have a close fit with our mathematical model. This illustrates that the manu-

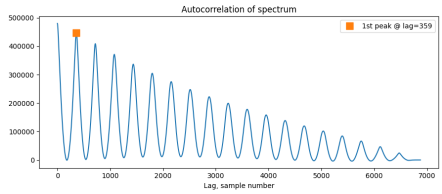

Fig. 11 MZI 2

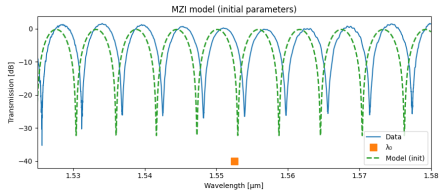

Fig. 12 MZI 1

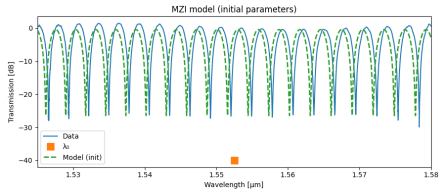

Fig. 13 MZI 2

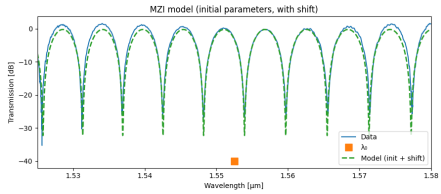

Fig. 14 MZI 1

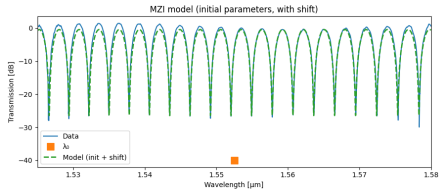

Fig. 15 MZI 2

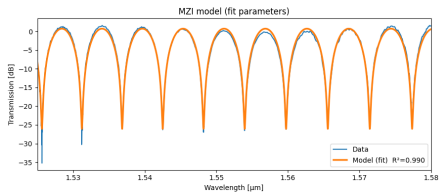

Fig. 16 MZI 1

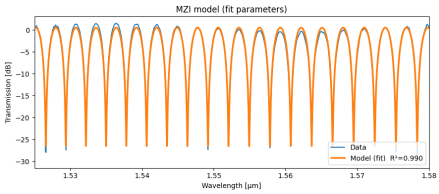

Fig. 17 MZI 2

factured dimensions are fairly consistent between these two waveguides, thus they fit the theoretical transfer function.

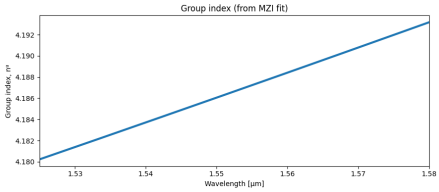

Fig. 18 MZI 1

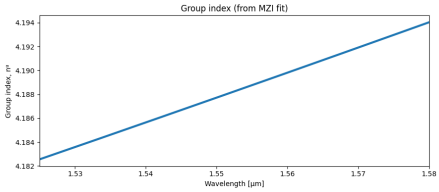

Fig. 19 MZI 2

Accounting for ΔL gives nearly identical graphs for group index, as we would expect. This aligns with the theory that n_g is a function of the cross sectional dimensions of the waveguide.

Now let us quickly compare the Scylla vs Iris results for MZI 1

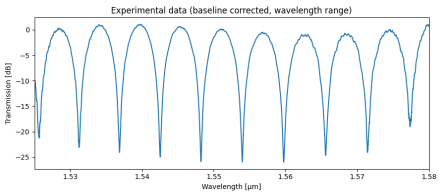

Fig. 20 Scylla

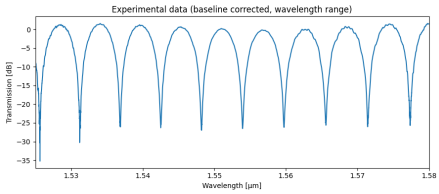

Fig. 21 Iris

The Scylla data has sharper bottoms for the null data. For Iris we have an R^2 of 0.9897, and for Scylla, we have an R^2 of 0.9826. Perhaps the shallower bottoms filter out noise resulting in a slightly higher R^2 value. Both methods are remarkably consistent overall.